

Admissible $\mathfrak{sl}(2)$ and $N = 2$, and their fusion product

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Joint work with

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Vertex operator algebras: basic intuition

Motivation

A vertex operator algebra (V, Y) is a complex vector space V together with a multiplication $Y : V \otimes_{\mathbb{C}} V \rightarrow V((z))$ satisfying some axioms that generalise **commutative unital associative algebras**.

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- V -tensor products (called fusion products).

Tensor categories from vertex operator algebras

The multiplication Y , actions Y_M and intertwining operators $\mathcal{Y}$ of V should be thought of as **two marked points** on the Riemann sphere, where the marked points are labelled by

- Y : two copies of V ,
- Y_M : V and M ,
- $\mathcal{Y}$: two modules.

More points can be added by composing $Y, Y_M, \mathcal{Y}$.

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- Changing charts in the stereographic projection of the Riemann sphere defines a duality structure (possibly weaker than rigid duals).

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- Creutzig and Ridout in the 10s noted that Kac-Wakimoto character formulae use analytic continuations and that in the larger category of weight modules some modules have characters which are the negatives of category $\mathcal{O}$ characters. Unlike category $\mathcal{O}$, the category of weight modules is neither finite nor semisimple.

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- Creutzig and Ridout proposed a generalisation of the Verlinde formula that equates characters with their series expansions and distinguishes these from analytic continuations. They used this generalised Verlinde formula to predict fusion product decomposition formulae (sometimes called fusion rules).

Affine $\mathfrak{sl}(2)$

Consider the $\mathfrak{sl}(2)$ Lie algebra with Cartan-Weyl basis $\{e, h, f\}$

$$[e, f] = h, \quad [h, e] = 2e, \quad [h, f] = -2f$$

and its affinisation $\widehat{\mathfrak{sl}}(2) = \mathfrak{sl}(2) \otimes \mathbb{C}[t, t^{-1}] \oplus \mathbb{C}K$

$$[x \otimes f(t), y \otimes g(t)] = [x, y] \otimes f(t)g(t) + \kappa(x, y) \operatorname{Res} f(t)g'(t)K, \quad x, y \in \mathfrak{sl}(2), f, g \in \mathbb{C}[t, t^{-1}].$$

Definition

Let $k \in \mathbb{C} \setminus \{-2\}$, then $V^k(\mathfrak{sl}(2)) = \operatorname{Ind}_{\mathfrak{sl}(2) \otimes \mathbb{C}[t] \oplus \mathbb{C}K}^{\widehat{\mathfrak{sl}}(2)} \mathbb{C}\Omega = U(\mathfrak{sl}(2) \otimes t\mathbb{C}[t])\Omega$, with $K = k \operatorname{id}$ is the unique vertex operator algebra defined by imposing

$$Y(x_{-1}\Omega, z)v = \sum_{m \in \mathbb{Z}} x_m v z^{-m-1}, \quad x_m = x \otimes t^m, \quad v \in V^k(\mathfrak{sl}(2)),$$

and continuing by normal ordering and derivatives.

Affine $\mathfrak{sl}(2)$

Proposition [Frenkel, Zhu, Kac, Gorelik, ...]

- ① $V^k(\mathfrak{sl}(2))$ satisfies the operator product relations

$$Y(x_{-1}\Omega, z)y_{-1}\Omega = \frac{k\kappa(x, y)\Omega}{z^2} + \frac{[x, y]_{-1}\Omega}{z} + \text{reg}, \quad x, y \in \mathfrak{sl}(2).$$

- ② The vector

$$\omega = \frac{1}{k+2} \left(\frac{1}{2}h_{-1}^2 + e_{-1}f_{-1} + f_{-1}e_{-1} \right) \Omega \in V^k(\mathfrak{sl}(2))$$

is a conformal vector. Its expansion $Y(\omega, z) = \sum_m L_m z^{-m-2}$ generates the Virasoro algebra at central charge

$$c_k = \frac{3k}{k+2}.$$

- ③ $V^k(\mathfrak{sl}(2))$ admits an ideal if and only if $\exists u, v \in \mathbb{Z}_{>0}$, $(u, v) = 1$, $u \geq 2$ and

$$k = \frac{u}{v} - 2.$$

Denote the simple quotient at these special levels by $A(u, v)$.

The rigidity of $A(u, v)$ weight modules

For $v = 1$, the $A(u, 1)$ are the famous rational $\mathfrak{sl}(2)$ Wess-Zumino-Witten modules. Here we are interested in $v > 1$.

Theorem [Nakano, Orosz Hunziker, Ros Camacho, SW; Creutzig, McRae, Yang]

Let $u, v \in \mathbb{Z}_{\geq 2}$, $(u, v) = 1$, then the category of weight modules, $A(u, v)\text{-Mod}^{\text{wt}}$, is rigid (actually even ribbon).

The $N = 2$ superconformal vertex algebra

Definition

Let $c \in \mathbb{C}$, and let V^c be the universal $N = 2$ superconformal vertex algebra. This is a vertex operator superalgebra with 4 strong generators:

- even: $\{J, L\}$, where J is a Heisenberg current and L is the conformal field.
- odd: $\{G^+, G^-\}$, conformal primaries of weight $\frac{3}{2}$ with respective J -weight $1, -1$.

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The coefficients in the expansion of the multiplication Y of $\{J, L, G^\pm\}$ generate the $N = 2$ superconformal Lie superalgebra. As a vector space

$$V^c = U(\{J_{-n}, L_{-n-1}, G_{-n-\frac{1}{2}}^\pm : n \in \mathbb{Z}_{>0}\})\Omega.$$

The $N = 2$ superconformal vertex algebra

Proposition [Kac, Gorelik]

V^c admits an ideal if and only if $\exists u, v \in \mathbb{Z}_{>0}$, $(u, v) = 1$, $u \geq 2$ and

$$c = 3 - \frac{6v}{u}.$$

Denote the simple quotient at these special central charges by $N(u, v)$.

The rigidity of $N(u, v)$ weight modules

For $v = 1$, the $N(u, 1)$ are the well known unitary rational $N = 2$ minimal models.

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Giving an outline of the proof of the rigidity of $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$ will be the goal for the rest of the talk.

Relating $A(u, v)$ and $N(u, v)$

Proposition [Gaberdiel, Eholzer]

- 1 There exists an embedding of vertex operator superalgebras

$$N(u, v) \otimes \mathbf{H} \hookrightarrow A(u, v) \otimes \mathbf{BC},$$

where $\mathbf{H}$ is the Heisenberg vertex algebra and $\mathbf{BC}$ is the fermionic ghost algebra.

- 2 For $v \geq 2$, $\mathbf{H}$ and $N(u, v)$ are each other's commutant.
- 3 Induction defines a tensor functor $N(u, v) \otimes \mathbf{H}\text{-Mod}^{\text{wt}} \rightarrow A(u, v) \otimes \mathbf{BC}\text{-Mod}^{\text{wt}}$, with restriction as its right adjoint.

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- 1 $\mathbf{H}$ and $\mathbf{BC}$ are free field algebras and so their categories of weight modules are very simple: $\mathbb{R}$ -graded vector spaces and super vector spaces, respectively.
- 2 Using that induction is a tensor functor means that $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$ can be easily related.

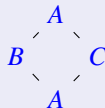
The categories of weight modules

An $A(u, v)$ - or $N(u, v)$ -module is weight if it decomposes into simultaneous h_0/J_0 eigenspaces and generalised L_0 eigenspaces, and these homogeneous spaces are all finite-dimensional. Denote the respective categories of weight modules by $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$.

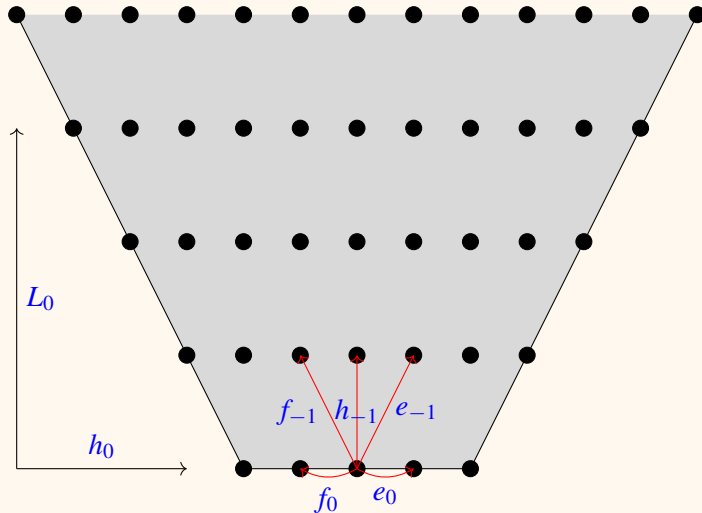
Proposition [Arakawa, Creutzig, Kawasetsu]

- ① The categories $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$ have sufficiently many injectives and projectives. All injectives are projective and vice versa.
- ② As linear categories $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$ are blockwise isomorphic.
- ③ There are two isomorphism classes of blocks.
 - ① Semisimple (come in continuous families).
 - ② Non-semisimple (finite number of families): Countably many simples.

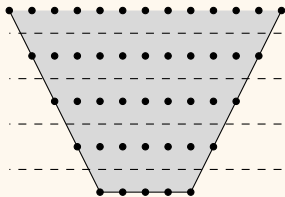
Indecomposable projectives/injectives have the form:



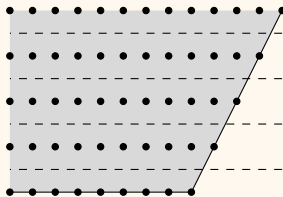
What simple $A(u, v)$ -modules look like



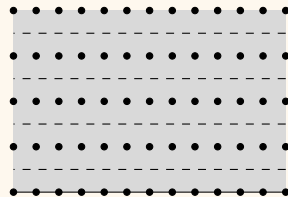
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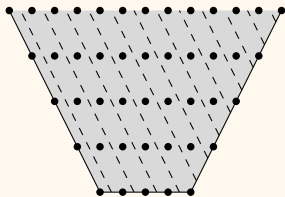


Highest weight

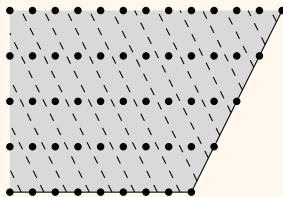


Dense

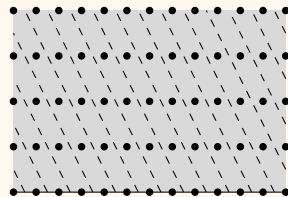
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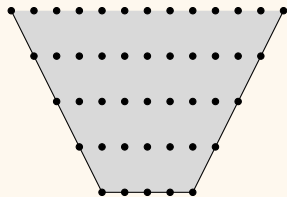


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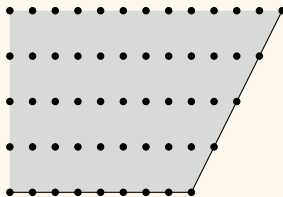


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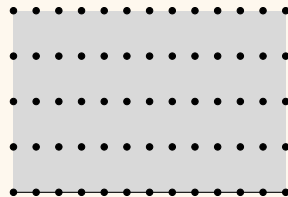
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$\sigma^\ell L_r$



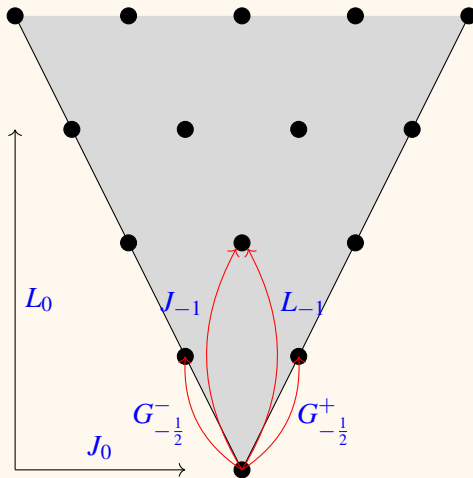
$\sigma^\ell D_{r,s}^+$



$\sigma^\ell E_{\lambda;r,s}$

- Here $\lambda \in \mathbb{R}/2\mathbb{Z}$, $1 \leq r \leq u-1$, $1 \leq s \leq v-1$.
- The $\sigma^\ell E_{\lambda;r,s}$ are simple outside of two special values.
- All simple $\sigma^\ell E_{\lambda;r,s}$ are projective and injective.
- The non-simple E -modules have two D -type composition factors.
- The D - and L -type modules are related by spectral flow.

What simple $N(u, v)$ -modules look like



Conjectured fusion rules

From a conjectured generalisation of the Verlinde formula Creutzig and Ridout predicted:

$$E_{\mu;r,s} \otimes E_{\mu';r',s'} = \bigoplus_{r'',s''} N_{r,r'}^{(u)r''} \left(N_{s,s'-1}^{(v)s''} + N_{s,s'+1}^{(v)s''} \right) E_{\mu+\mu';r'',s''} \\ \oplus \bigoplus_{r'',s''} N_{r,r'}^{(u)r''} N_{s,s'}^{(v)s''} \left(\sigma^{-1} E_{\mu+\mu'+\frac{u}{v};r'',s''} \oplus \sigma E_{\mu+\mu'-\frac{u}{v};r'',s''} \right),$$

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The L_r were shown to be rigid by [Creutzig, Huang, Yang].

The free field realisation of $A(u, v)$

Proposition [Adamović; Semikhatov]

Let $v \geq 2$, and let $M(u, v)$ be the Virasoro minimal model vertex operator algebra at central charge $c = 1 - 6 \frac{(u-v)^2}{uv}$.

- 1 There exists an embedding of vertex operator algebras

$$A(u, v) \hookrightarrow M(u, v) \otimes \mathbf{L},$$

where $\mathbf{L}$ is the rank 2 Heisenberg algebra extended along a rank 1 lattice (with trivial lattice pairing).

- 2 There exists a screening current $S(z)$ such that the image of $A(u, v)$ is the kernel of $S = \text{Res } S(z)$.

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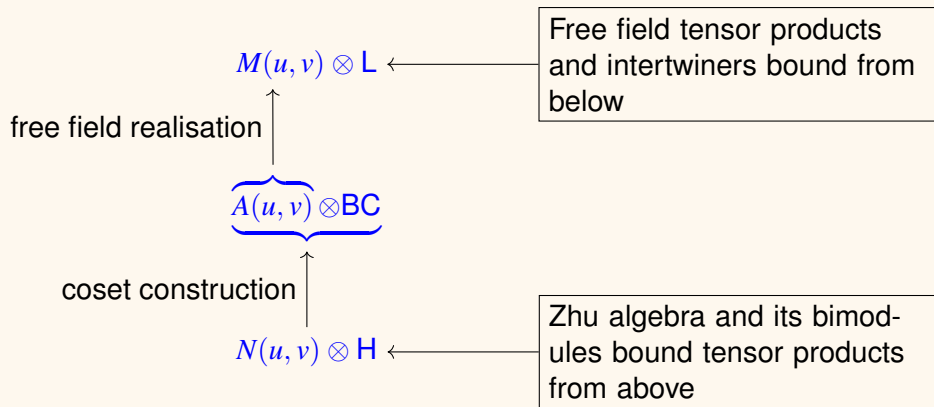
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 $S(z)$ is associated with the “ $(1, 2)$ -module of $M(u, v)$ ”

Bounding $A(u, v)$ from above and below



Bounding fusion rules from above

- 1 The conformal weights of $N(u, v)$ are always bounded below. They are hence amenable to Zhu theory.
- 2 Zhu theory gives relations/constraints on fusion rules, but in practice it is very hard to prove when one has all relations.
- 3 The $N(u, v)$ counterparts to the $E_{\mu;1,1}$ have a singular vector at degree 1 that can be expressed in simple formulae.
- 4 Using the above it is easy to show that the $E_{\mu;1,1}$ fusion rules cannot be larger than Creutzig and Ridout's conjecture.

Bounding fusion rules from below

- 1 Free field algebras come with easy intertwining operators (multilinear maps).
- 2 The images of these intertwining operators are then quotients of the fusion product for the subalgebra (i.e. lower bounds).
- 3 Here the free field intertwining operators give the spectrally flowed summands of the conjectured fusion rules.
- 4 For the unflowed summands we needed to develop a new method of constructing intertwining operators by twisting the free field intertwining operators by the screening operators.

The rigidity of the $E_{\lambda;1,1}$

Definition

An object X admits a rigid dual $X^\vee$, if there exist

$$\mathrm{ev}_X : X^\vee \otimes X \rightarrow \mathbf{1}, \quad \mathrm{coev}_X : \mathbf{1} \rightarrow X \otimes X^\vee$$

such that

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{coev}_X \otimes \mathrm{id}_X} & X \otimes X^\vee \otimes X \\ & \searrow \mathrm{id}_X & \nearrow \\ & X & \end{array} \quad \begin{array}{ccc} X^\vee & \xrightarrow{\mathrm{id}_{X^\vee} \otimes \mathrm{coev}_X} & X^\vee \otimes X \otimes X^\vee \\ & \searrow \mathrm{id}_{X^\vee} & \nearrow \\ & X^\vee & \end{array}$$

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 & & & & \\
 X^\vee & \xrightarrow{\mathrm{id}_{X^\vee} \otimes \mathrm{coev}_X} & X^\vee \otimes X \otimes X^\vee & \xrightarrow{\mathrm{ev}_X \otimes \mathrm{id}_{X^\vee}} & X^\vee \\
 & \curvearrowright & & & \\
 & \mathrm{id}_{X^\vee} & & &
 \end{array}
 \end{array}$$

- The conjectured rigid dual of $\sigma^\ell E_{\lambda;1,1}$ was $\sigma^{-\ell} E_{-\lambda;1,1}$.
- We needed to determine $E_{-\lambda;1,1} \otimes E_{\lambda;1,1} = \sigma^{-1} P_{u-1,v-1} \oplus (1 - \delta_{v,2}) E_{0;1,2}$.
- $\sigma^{-1} P_{u-1,v-1}$ is the projective cover of $\mathbf{1} = A(u, v)$.
- To construct ev and coev one therefore needs an intertwining operator of type $\begin{pmatrix} \sigma^{-1} P_{u-1,v-1} \\ E_{-\lambda;1,1}, E_{\lambda;1,1} \end{pmatrix}$.

The rigidity of the $E_{\lambda;1,1}$

- Following [Adamović], $\sigma^{-1}P_{u-1,v-1}$ can be constructed as a sum of free field modules with the free field action twisted by screening operators.
- We constructed an intertwining operator of type $\left(\begin{smallmatrix} \sigma^{-1}P_{u-1,v-1} \\ E_{-\lambda;1,1}, E_{\lambda;1,1} \end{smallmatrix} \right)$ by twisting a free field intertwining operator by screening operators.
- This is the first example in the literature of an explicit logarithmic intertwining operator.

The rigidity of all weight modules

The rigidity of all weight modules then follows inductively from the following:

- Rigid times rigid is rigid.
- Direct summands of rigid modules are rigid.
- Under some assumptions (which apply here), if a short exact sequence has two rigid terms, then the third is too.

Summary

- The monoidal categories $A(u, v)\text{-Mod}^{\text{wt}}$ and $N(u, v)\text{-Mod}^{\text{wt}}$ are rigid.
- This was shown independently by [Creutzig, McRae, Yang] and [Nakano, Orosz Hunziker, Ros Camacho, SW].
- The approach of [Creutzig, McRae, Yang] uses algebra extensions and Grothendieck-Verdier duality. Looks promising for higher rank algebras provided certain key products of modules can be computed explicitly.
- Our approach used Zhu theory (well studied) and novel screening operator methods. These in particular help with computing fusion products explicitly.

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Thank you!